PREDICTIVE ANALYTICS

Group Project

Group Project Brief and Assessment Rubric

Item	Specification
Assessment	Group Project
Mode	Collaborative project with shared analytical responsibility
Project themes	Finance and Business; Agriculture and Food Systems; Energy and Environment; Public Services and Mobility
Project options	Regression, classification, or time-series forecasting
Presentation	Meetings 14-15; evidence-based business pitch; 12 minutes plus 5 minutes stakeholder-led discussion and Q&A
Submission	One ZIP package submitted through RISE by 19 October 2026 at 20.00 WIB
Technical interview	26 October 2026; lecturer assessment of methodological understanding and individual ownership

Purpose

This project assesses each group's ability to complete an end-to-end predictive analytics workflow and extend it through systematic hyperparameter tuning, evidence-based model selection, clear interpretation, actionable insights, and relevant business or organizational implications. The project builds on the individual midterm project while requiring a new dataset, collaborative analysis, and a formal presentation.

Member and Dataset registration: s.id/listdatapred02

1 Project Task

Select one real, public, and verifiable dataset within one of the four approved themes. Each group may freely define its prediction problem and choose regression, classification, or time-series forecasting, provided that the target and evaluation design are appropriate.

The exact dataset used by each group must be different. Groups may work within the same broad theme, but they may not use the same dataset, even when proposing a different model or target. A dataset or notebook used for the individual midterm project may not be reused.

2 Intended Learning Outcomes

- Formulate a clear predictive question, target variable, unit of analysis, and stakeholder context.
- Assess data quality and perform justified data preparation, exploratory analysis, and feature engineering.
- Develop valid baselines and multiple candidate models using an evaluation strategy that prevents data leakage.
- Tune selected models systematically and compare default and tuned performance fairly.
- Interpret the final model and distinguish predictive evidence from causal claims.
- Translate analytical findings into specific, evidence-based recommendations and business or organizational implications.
- Communicate the project clearly as a team and demonstrate ownership of the analysis.

3 Project Themes

Theme	Possible areas of application
Finance and Business Analytics	Financial risk, insurance, customer behavior, marketing, sales, revenue, or business demand
Agriculture and Food Systems	Crop production, agricultural risk, commodity prices, food demand, livestock, or supply systems
Energy and Environment	Energy consumption, electricity demand, renewable generation, air quality, weather, or environmental indicators
Public Services and Mobility	Transport, education, health-service demand, tourism, public operations, or service performance

The examples are illustrative rather than fixed topics. Groups are responsible for defining a feasible and meaningful prediction problem within the selected theme.

4 Dataset Selection and Registration

Before beginning the full analysis, every group must list its proposed dataset through the course dataset registry.

Dataset registration: s.id/listdatapred02

The registration entry should include:

- Group number or group name and member names.
- Selected project theme and proposed project title.
- Prediction type and intended target variable.
- Dataset title, publisher or owner, source link, and access date.
- Number of observations, potential predictors, and time coverage where applicable.

If two groups register the same dataset, one group will be required to select a different dataset. Listing a dataset records the group's choice and supports the uniqueness check; it does not remove the group's responsibility to confirm that the dataset satisfies all requirements below.

5 Dataset Selection Requirements

Requirement	Minimum expectation
Public and verifiable	Use a credible public or official source such as data.go.id, BPS, UCI, World Bank, or another traceable repository.
Source citation	Provide the dataset title, publisher or owner, source link, access date, and original source when a secondary repository is used.
Clear target	Define a meaningful target that matches regression, classification, or time-series forecasting.
Sufficient size	For tabular projects, use at least 1000 observations and five potential predictors. For time series, use at least 100 ordered observations unless otherwise approved.
Requires preparation	The dataset should contain realistic needs for cleaning, transformation, feature engineering, or analytical judgment.
Authentic and ethical	Do not use synthetic, confidential, personally identifying, or unlawfully obtained data.
Original and unique	Do not reuse a midterm dataset and do not use the same dataset as another group.
Computationally feasible	The complete workflow should run within a reasonable time in Google Colab or an equivalent environment.

6 Required Submission Package

Submit one compressed folder through the course RISE no later than 19 October 2026 at 20.00 WIB using the following naming convention:

GroupNumber_ShortProjectTitle_PredictiveAnalytics.zip

Required file	Expected format and content
Jupyter Notebook	.ipynb file containing the complete reproducible analysis and visible outputs
Dataset	The exact dataset file or files used by the notebook in a portable format such as CSV or XLSX
Presentation slides	PPTX or PDF version of the final group presentation
Contribution statement	A concise record of each member's analytical, coding, interpretation, and presentation contributions

7 Jupyter Notebook Requirements

The notebook must be organized as a reproducible analytical report. Markdown cells should explain the reasoning, methodological choices, and interpretation of outputs. Code output without explanation is not sufficient.

No	Required section	Minimum content
1	Project overview	Context, stakeholder, predictive question, task type, target, and unit of analysis
2	Dataset description	Source, dimensions, time coverage, observation unit, target, predictors, and relevant data dictionary
3	Data-quality assessment	Data types, missing values, duplicates, invalid values, outliers, target distribution, and leakage risks
4	Data preparation	Documented cleaning, type correction, missing-value handling, outlier decisions, and reproducible preprocessing
5	Exploratory data analysis	Focused univariate and multivariate analysis that informs modeling choices
6	Feature engineering	Relevant transformations, encoding, scaling, time features, aggregation, or feature selection with rationale

7 Jupyter Notebook Requirements Continued

No	Required section	Minimum content
7	Baseline model	A simple benchmark establishing the minimum performance that trained models should improve upon
8	Candidate models	Required model families, correct split logic, reproducibility controls, and concise rationale
9	Hyperparameter tuning	Models selected for tuning, method, search space, validation design, metric, and best parameters
10	Model evaluation	Default-versus-tuned comparison, appropriate metrics, diagnostics, and final model selection
11	Model interpretation	Coefficients, feature importance, or another appropriate method with accurate, non-causal interpretation
12	Actionable insights	Evidence-based findings connected to specific actions that a relevant stakeholder could consider
13	Business implications	Practical relevance, expected organizational use, implementation considerations, limitations, and next steps

8 Minimum Modeling Requirements

Project type	Required minimum models	Recommended evaluation evidence
Regression	Mean or median baseline; linear regression; at least two additional models, including a tree-based or ensemble approach	MAE and RMSE; R^2 as supporting evidence; residual or error analysis
Classification	Dummy baseline; logistic regression; at least two additional models, including a tree-based or ensemble approach	Confusion matrix; precision, recall, and F1-score; accuracy or ROC-AUC where appropriate
Time-series forecasting	Naive or seasonal-naive baseline; at least three forecasting approaches	Chronological evaluation; MAE or RMSE; MAPE only when valid; forecast plots and error analysis

9 Hyperparameter Tuning and Model Selection

After the initial model comparison, each group must select the two strongest non-baseline models for systematic hyperparameter tuning.

- Tune at least two relevant hyperparameters for each selected model.
- Use GridSearchCV, RandomizedSearchCV, Optuna, or another clearly justified search method.
- Use cross-validation appropriate to the task. Time-series projects must use chronological, rolling, or expanding-window validation.
- State the tuning metric and ensure it is consistent with the primary model-selection objective.
- Report the search space, best parameters, default performance, tuned performance, and observed improvement or trade-off.
- Do not use the final test data to choose features, tune parameters, or select the final model.

Hyperparameter tuning is not required to produce a large improvement. A valid conclusion may be that tuning provides only a small gain or that a simpler model remains preferable.

10 Validation and Technical Requirements

- Use train, validation, and test logic appropriate to the analytical task.
- Fit preprocessing only on training data and use pipelines where appropriate to prevent leakage.
- Compare models using the same evaluation split or folds and the same primary metric.
- For imbalanced classification, report class-sensitive metrics and explain the implications of prediction errors.
- For time series, preserve temporal order, define the forecast horizon, and never shuffle observations.
- Set random seeds where applicable and keep the workflow computationally reasonable.

11 Interpretation Actionable Insights and Business Implications

The final analysis must move beyond reporting metric values. Each group should identify at least three important findings and translate them into at least two feasible recommendations. Financial calculations are not required when credible cost data are unavailable, and feature importance must not be presented automatically as causal influence.

Element	Expected evidence
Finding	A clear pattern, predictor, error behavior, or model result supported by analysis
Evidence	Relevant metric, visualization, feature-importance result, coefficient, or diagnostic
Actionable insight	A specific action that a stakeholder could consider based on the evidence
Business implication	How the finding may influence planning, prioritization, monitoring, service delivery, or operational decisions
Limitation	Data, validation, generalization, interpretation, or implementation constraints

12 Reproducibility Standard

The submitted notebook must run successfully using Restart Runtime then Run All.

- No execution error, missing variable, hidden manual step, or dependency on a previous session.
- Use relative paths or one clearly defined data path; do not use a personal local-computer path.
- Include all imports and installation commands at the beginning only when necessary.
- Include the exact dataset version used to generate the reported results.
- Ensure tables, plots, labels, legends, and units are readable and consistent.

13 Group Presentation

The nine groups will present across Meetings 14 and 15. Four groups will present in Meeting 14 and five groups in Meeting 15. The presentation order and designated stakeholder group are fixed in the schedule below. All groups must submit the complete final package by 19 October 2026 at 20.00 WIB, before the first presentation meeting.

Component	Expectation
Presentation	Maximum 12 minutes using approximately 10-12 focused slides. Present the work as an evidence-based business pitch to relevant organisational decision-makers.
Questions and answers	Approximately 5 minutes, beginning with questions from the designated C-level stakeholder group and followed by the lecturer or audience.
Team participation	Every member must contribute meaningfully and be prepared to explain the group's analysis, evidence, and recommendations.
Demonstration	Open the notebook or relevant outputs when requested. A separate application or dashboard is not required.
Pitch objective	Convince the stakeholder that the analysis is reliable and that the proposed recommendation is relevant, feasible, and valuable, while acknowledging limitations and implementation risks.

Presentation Format and Stakeholder Role

Groups may determine their own slide sequence and storytelling approach. The presentation should not simply reproduce the notebook step by step. It should function as a concise business pitch in which technical evidence supports a clear recommendation. The audience should understand the problem, why the results are credible, what action is proposed, and what limitations should be considered.

For every presentation, one other group will act as the stakeholder panel. The panel should assume an appropriate C-level role, such as Chief Executive Officer, Chief Financial Officer, Chief Risk Officer, Chief Marketing Officer, Chief Operating Officer, Chief Data Officer, or another senior decision-maker relevant to the case.

Before the session, the stakeholder group should review the presenting group's project title and business context. During the Q&A, it should challenge the reliability, feasibility, risks, and organisational value of the recommendation. The stakeholder group is not expected to repeat technical questions already answered in the presentation.

Presentation and Stakeholder Schedule

Meeting	Presentation order	Presenting group	C-level stakeholder group
Meeting 14	1	Group 1	Group 9
Meeting 14	2	Group 2	Group 8
Meeting 14	3	Group 3	Group 7
Meeting 14	4	Group 4	Group 6
Meeting 15	1	Group 5	Group 1
Meeting 15	2	Group 6	Group 2
Meeting 15	3	Group 7	Group 3
Meeting 15	4	Group 8	Group 4
Meeting 15	5	Group 9	Group 5

Each group therefore serves once as the presenting team and once as the designated stakeholder panel. The five-minute discussion period includes the stakeholder panel's questions; it does not add extra time to the presentation slot.

14 Technical Interview

The technical interview will be conducted by the lecturer on 26 October 2026 (Meeting 15). Its purpose is to verify each member's understanding of the submitted work and to assess individual ownership of the group analysis. The detailed time allocation will be announced by the lecturer.

Interview area	Expected evidence
Problem and data	Explain the target, unit of analysis, dataset suitability, data-quality issues, and important preparation decisions.
Model development	Explain the baseline, candidate models, validation design, metric selection, and leakage-prevention steps.
Hyperparameter tuning	Explain the selected models, search space, tuning method, best parameters, and default-versus-tuned comparison.
Results and interpretation	Interpret the final model, diagnostics, important predictors or patterns, and limitations without unsupported causal claims.
Recommendation	Connect analytical evidence to the proposed action and explain its expected organisational value, feasibility, and risks.

The lecturer may direct different questions to different members. Every member should understand the complete analytical workflow, not only the section listed in the contribution statement. Inability to explain key choices or outputs may reduce the individual ownership and technical-interview result.

15 Assessment Rubric 100 Points

The project is graded using evidence in the submitted package, the business-pitch presentation, the designated stakeholder role, and the technical interview. Full-credit evidence is described below.

Criterion	Pts	Evidence expected for full credit
1 Problem formulation and dataset suitability	10	Clear question, target, unit, stakeholder, theme, registration, source, ethics, feasibility, and dataset uniqueness
2 Data quality assessment and preparation	15	Systematic audit; justified handling of missing values, duplicates, types, invalid values, and outliers; reproducible and leakage-safe steps
3 Exploratory analysis and feature engineering	10	Focused EDA informs modeling; meaningful features and correct transformations are explained
4 Baseline and candidate-model development	15	Valid baseline and all required models; correct preprocessing and split logic; justified choices
5 Validation hyperparameter tuning and selection	20	Appropriate validation; transparent search space; fair default-versus-tuned comparison; final model selected using evidence
6 Interpretation insights and implications	15	Sound interpretation; specific actionable insights; relevant business or organizational implications; honest limitations; no unsupported causal claims
7 Reproducibility and code quality	5	Restart Runtime then Run All succeeds; portable paths; organized code; included files; readable outputs
8 Presentation teamwork and ownership	10	Clear and convincing business pitch; balanced participation; evidence-based recommendation; meaningful stakeholder-panel contribution; accurate explanations; and demonstrated individual ownership in the technical interview
TOTAL	100	The score is converted to the applicable group-task and presentation contribution in the course assessment plan

How Partial Credit Is Determined

Performance level	General interpretation
Complete and convincing	Correct, fully documented, justified, reproducible, tuned appropriately, and interpreted accurately
Mostly complete	Core work is correct, but explanation, tuning evidence, diagnostics, interpretation, or polish is incomplete
Partially demonstrated	Some relevant work is present, but important errors, omissions, or weak justification limit confidence
Not demonstrated	Required evidence is absent, unusable, substantially incorrect, or cannot be explained

Important Grading Implications

- An unregistered, unverifiable, reused, or duplicated dataset may be rejected and must be replaced.
- A notebook that does not complete Restart Runtime then Run All successfully cannot receive full reproducibility marks.
- Missing baselines, required candidate models, or required tuning evidence reduces the relevant criterion score.
- Target leakage, train-test contamination, tuning on test data, or inappropriate time-series splitting is a major methodological error.
- Unsupported causal statements, copied interpretations, or inability to explain the work reduces interpretation and ownership marks.
- Meeting the minimum model count or tuning requirement alone does not guarantee a high score; reasoning, validation, interpretation, and communication are assessed.

Submission Checklist

1. Jupyter Notebook with visible outputs.
2. Exact dataset file or files used in the analysis.
3. Presentation slides in PPTX or PDF format, prepared as an evidence-based business pitch.
4. Contribution statement listing each member's role and outputs.
5. All required files compressed into one ZIP folder and uploaded through RISE.

Final check: Before submission, confirm that the dataset has been listed at s.id/listdatapred02 and that no other group is using the same dataset.